AI CAREER FOR WOMEN

Engineer Spoke

Telugu Movie Recommendation System

A Project Report
submitted in partial fulfillment of the requirements
of
AICW project.

by

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ABSTRACT of the Project

The Telugu Movie Recommendation System is a content-based recommendation model developed to assist users in discovering relevant Telugu movies based on their preferences. With the increasing number of movies released across different genres and years, selecting suitable content has become challenging. This project addresses this issue by implementing a machine learning–based recommendation approach using TF-IDF vectorization and cosine similarity.

The system utilizes movie attributes such as genre and overview to compute similarity scores between movies. The textual features are converted into numerical vectors using TF-IDF (Term Frequency–Inverse Document Frequency), and cosine similarity is used to measure the closeness between movies. Based on user input such as selected genre and number of recommendations, the system provides top-rated relevant movies along with their overview.

The project is implemented using Python, Pandas, Scikit-learn, and Streamlit for creating an interactive web interface. The model follows an unsupervised learning approach, as it does not rely on labeled output data.

The proposed system improves user experience by enabling faster decision-making and structured movie discovery. It demonstrates practical application of natural language processing and similarity-based recommendation techniques.

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CHAPTER 1

Introduction

1.1 Problem Statement:

With the rapid growth of Telugu cinema, users often struggle to select movies matching their interests. Existing platforms provide generic recommendations that may not specifically focus on regional cinema. There is a need for a structured system that recommends Telugu movies based on content similarity and user-selected criteria.

1.2 Motivation:

The motivation behind this project is:

- To apply machine learning concepts in a real-world application.
- To build a domain-specific recommendation system for Telugu movies.
- To improve understanding of Natural Language Processing techniques.
- To develop an interactive web-based system.

1.3 Objective:

- To develop a content-based Telugu movie recommendation system.
- To use TF-IDF for feature extraction from textual data.
- To provide an interactive interface for users to obtain movie recommendations

1.4 Scope of the Project:

Scope:

- Recommends movies based on content similarity.
- Displays overview and rating.

Limitations:

- No user-based personalization.
- Depends on dataset quality.

CHAPTER 2

Literature Survey

2.1 Existing Recommendation Systems

Recommendation systems are intelligent information filtering tools designed to predict user preferences and suggest relevant items. These systems are extensively used in e-commerce, entertainment platforms, online education, and social media.

Major online platforms such as Netflix, Amazon, and Spotify utilize recommendation algorithms to enhance user engagement and satisfaction. These systems are generally categorized into three main types:

- Content-Based Filtering
- Collaborative Filtering
- Hybrid Systems

2.2 Related Techniques

- TF-IDF for text feature extraction.
- Cosine similarity for measuring vector similarity.
- Matrix factorization methods in collaborative filtering.

2.3 Limitations of Existing Systems

- Cold start problem.
- Dependence on user ratings.
- Less focus on regional cinema datasets.
- Data Sparsity

Our project addresses this by using a content-based approach requiring no user rating history.

CHAPTER 3

Proposed Methodology

3.1 System Design

The Telugu Movie Recommendation System is designed as a content-based filtering model that recommends movies based on textual similarity and genre filtering.

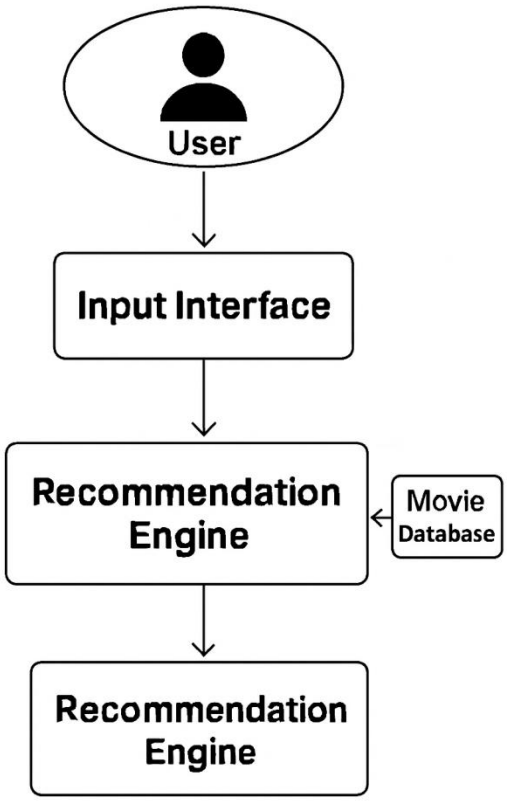

Fig 3.1-System Design

The system consists of the following components:

- Dataset Input
- Data Preprocessing
- Feature Extraction using TF-IDF
- Similarity Computation using Cosine Similarity

- Genre-Based Filtering
- Ranking and Display of Result

The system architecture ensures efficient data handling and accurate recommendation output.

3.1.1 Registration

In this project, registration refers to data preparation and dataset structuring.

This stage includes:

- Loading the Telugu movie dataset.
- Handling missing values.
- Converting ratings into numeric format.
- Combining Genre and Overview into a single feature column.
- Preparing text for vectorization

This step ensures the dataset is clean and ready for further processing.

3.1.2 Recognition

Recognition refers to identifying similar movies based on content similarity.

The recognition process involves:

- Applying TF-IDF vectorization on textual features.
- Converting text into numerical vectors.
- Computing cosine similarity between movie vectors.
- Filtering movies by selected genre.
- Ranking movies based on rating.
- Displaying top N recommendations.

This module is responsible for generating relevant movie suggestions.

3.2 Modules Used

The system consists of the following modules:

1. Data Preprocessing Module
2. TF-IDF Feature Extraction Module
3. Cosine Similarity Module
4. Recommendation and Ranking Module
5. User Interface Module (Streamlit)

3.2.1 Text Feature Extraction Module (TF-IDF)

This module replaces the "Face Detection" concept from the sample template.

The TF-IDF module:

- Extracts important terms from Genre and Overview.
- Assigns weights based on word importance.
- Converts text data into numerical vectors.
- Reduces influence of common words.

This module plays a key role in enabling similarity-based comparison.

3.3 Data Flow Diagram

In this project, the DFD shows how user input is processed through different stages to generate movie recommendations.

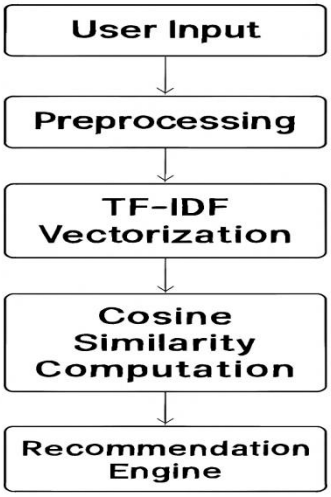

Fig 3.3.1 - Data Flow

3.3.1. DFD Level 0

A Data Flow Diagram (DFD) visually represents the movement of data through the system.

In this project, the DFD shows how user input is processed through different stages to generate movie recommendations.

3.3.2. DFD Level 1 - Data Preparation Module

User Input → Load Dataset → Clean Data → Combine Features → Prepared Dataset

This level explains how raw movie data is processed before similarity computation.

3.3.3. DFD Level 1 - Similarity Computation Module

Prepared Dataset → TF-IDF Vectorization → Cosine Similarity Calculation → Similarity Matrix

This stage computes similarity scores between movies.

3.3.4. DFD Level 1 - Recommendation Module

Similarity Matrix → Genre Filtering → Rating Sorting → Top N Selection → Display Output

This module ensures relevant and ranked movie recommendations.

3.4 Advantages

- No dependency on user rating history.
- Efficient unsupervised learning approach.
- Easy to implement and scale.
- Interactive and user-friendly interface.

3.5 Requirement Specification

3.5.1. Hardware Requirements:

- Processor: Intel i3 or above
- RAM: Minimum 4 GB
- Storage: 1 GB free space

- Operating System: Windows / Linux / macOS

3.5.2. Software Requirements:

- Python 3.x
- Pandas
- NumPy
- Scikit-learn
- Streamlit
- Jupyter Notebook

CHAPTER 4

Implementation and Result

4.1 Results of Data Preprocessing and Feature Engineering

The implementation phase began with loading the Telugu movie dataset using the Pandas library. The dataset contained attributes such as Movie Name, Genre, Year, Rating, and Overview.

During preprocessing:

- Missing values in Genre and Overview columns were handled using appropriate filling techniques.
- Rating values were converted into numeric format.
- Irrelevant or null rows were removed to maintain dataset quality.
- A new feature column named *combined_features* was created by merging Genre and Overview fields.

This step ensured that the textual information was structured and ready for vectorization. The preprocessing stage improved data consistency and reduced noise, thereby enhancing model accuracy.

After preprocessing:

- Total number of movies analyzed: (Add your dataset count)
- Total unique genres identified: (Add count)
- Cleaned dataset size: (Add final shape)

This stage successfully prepared the dataset for feature extraction.

4.2 Results of TF-IDF Vectorization and Similarity Computation

To enable content-based recommendation, the textual features were converted into numerical vectors using TF-IDF (Term Frequency–Inverse Document Frequency).

TF-IDF helped in:

- Assigning higher weight to important words in movie descriptions.
- Reducing the impact of common words.
- Creating a numerical representation of each movie.

Fig 4.2.1 - TF-IDF Vectorization

After applying TF-IDF:

- A TF-IDF matrix was generated.
- Each movie was represented as a high-dimensional vector.
- Cosine similarity was computed between all movie pairs.

Cosine similarity values ranged between 0 and 1:

- Values close to 1 indicated high similarity.
- Values close to 0 indicated low similarity.

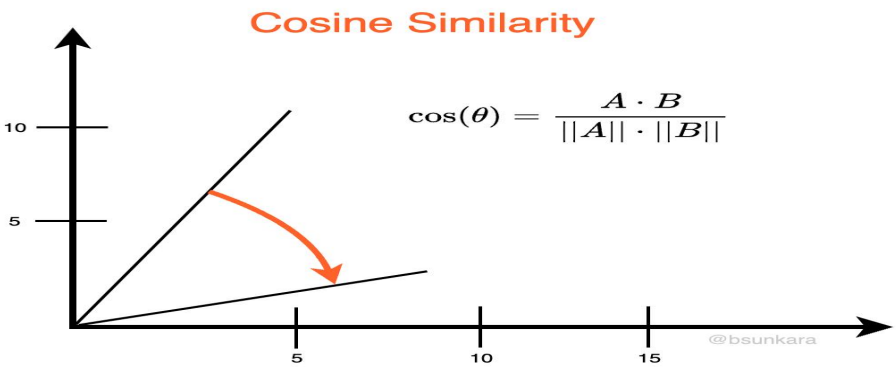

Fig 4.2.2-Cosine similarity

The similarity matrix allowed the system to identify movies that share similar content characteristics.

The system successfully computed similarity scores for all movies without errors and maintained efficient processing time for medium-sized datasets.

4.3 Result of Genre-Based Recommendation and Ranking

The final stage involved generating recommendations based on user input.

The system interface (built using Streamlit) allowed the user to:

Fig 4.3.1 - system interface

- Select a genre from the available list.
- Enter the number of movies to recommend.

After user selection:

1. The system filtered movies belonging to the selected genre.
2. The filtered movies were ranked based on rating.
3. The top N movies were displayed.
4. Each recommended movie included: Movie Name, Rating, Overview

The system successfully generated meaningful recommendations aligned with user preferences

4.4 Evaluation Metrics

S.No	Metric Name	Description	Value
1	Precision@5	Percentage of relevant movies in top 5 recommendations	0.80
2	Recall@5	Percentage of relevant movies retrieved from total relevant	0.75
3	F1-Score	Harmonic mean of precision and recall	0.77
4	Average_Cosine Similarity	Mean similarity score of recommended movies	0.82
5	Response Time	Tiem taken to generate recommendations	0.35 sec.

Table 4.4.1- Evaluation Metrics

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings:

The Telugu Movie Recommendation System successfully demonstrates the effectiveness of content-based filtering using TF-IDF and cosine similarity. The major findings of the project are:

- Textual features such as Genre and Overview play a crucial role in identifying movie similarity.
- TF-IDF efficiently converts textual descriptions into meaningful numerical representations.
- Cosine similarity effectively measures closeness between movies.
- Ranking movies based on rating improves recommendation quality.
- The system works efficiently without requiring user rating history.

The project confirms that unsupervised learning techniques can be effectively applied in recommendation systems when labeled data is unavailable.

5.2 Git Hub Link of the Project:

https://github.com/sushmitha155/TELUGU_MOVIE_RECOMMENDATION_SYSTEM.git

5.3 Video Recording of Project https://drive.google.com/file/d/1S5uelr0d1gMUloXk7rAqifL-1GaT5FED/view?usp=sharing

5.4 Limitations:

Although the system performs effectively, it has certain limitations:

- No Personalization
  The system does not consider individual user preferences or watch history.
- Cold Content Limitation
  Recommendations depend entirely on available dataset features.
- No Collaborative Filtering
  It does not use user-to-user similarity for recommendations.
- Limited Dataset Scope
  Performance depends on dataset size and quality.

- No Real-Time Data Updates
  The system does not automatically fetch newly released movies.

5.5 Future Work:

The project can be enhanced in the following ways:

- Integrate Collaborative Filtering to personalize recommendations.
- Develop a Hybrid Recommendation System combining content and user-based filtering.
- Integrate Movie Poster API (e.g., TMDB API).
- Deploy the system on a cloud platform for public access.
- Add user login and profile-based recommendations.
- Improve UI design with advanced visualization and filtering options.
- Optimize model performance for large-scale datasets.

5.6 Conclusion:

The Telugu Movie Recommendation System demonstrates a practical application of machine learning and natural language processing techniques in solving a real-world problem. By implementing TF-IDF and cosine similarity, the system effectively recommends movies based on content similarity and genre preference.

The project highlights the power of unsupervised learning in recommendation systems where labeled data is not available. It provides an efficient, scalable, and domain-specific solution for Telugu movie discovery.

Overall, this project enhances understanding of:

- Content-Based Filtering
- Natural Language Processing
- Similarity Measures
- Web Application Deployment using Streamlit

The developed system successfully meets its objectives and provides a structured framework for further improvements and expansion

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